DIWALI SALES STORE

Project overview

The sales Dataset from Diwali Sales Store with aim of deeply understanding the business operations through analysis of sales records, know major customers and the highest revenue generating products with aim of increasing the Revenue and guide in making strategic business decisions.

Dataset Summary

Columns: 15

Rows: 11251

Key features:

- Customer demographics: Gender, Age group, Age, Marital status, State, zone.
- Purchase details: Product ID, occupation, product category, orders, Amount.

Exploratory Data Analysis with Python

Started with importing data using Pandas.

- Data cleaning using df.info() to check the contents of the data and df. describe() to check statistics of the data.
- Removed the empty columns from the dataset: Status and unnamed1
- I fixed the null values in Amount column

```
df.info()
```

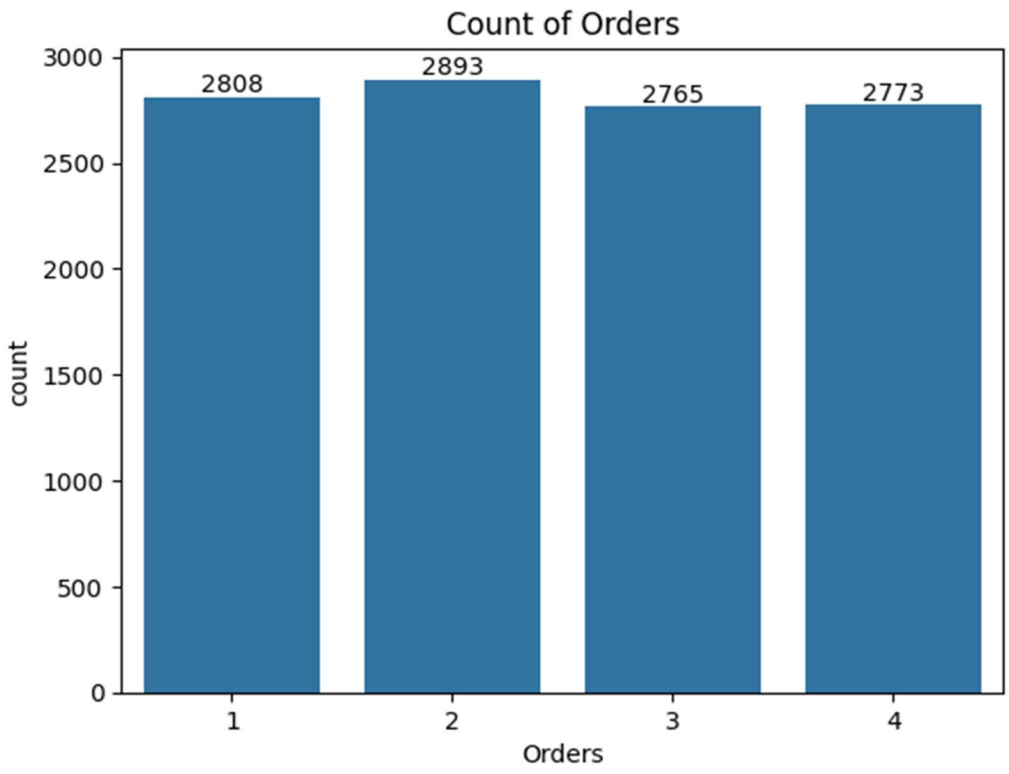
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11251 entries, 0 to 11250
Data columns (total 15 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   User_ID           11251 non-null  int64  
 1   Cust_name         11251 non-null  object 
 2   Product_ID        11251 non-null  object 
 3   Gender            11251 non-null  object 
 4   Age Group         11251 non-null  object 
 5   Age               11251 non-null  int64  
 6   Marital_Status    11251 non-null  int64  
 7   State             11251 non-null  object 
 8   Zone              11251 non-null  object 
 9   Occupation        11251 non-null  object 
 10  Product_Category  11251 non-null  object 
 11  Orders            11251 non-null  int64  
 12  Amount            11239 non-null  float64
 13  Status            0 non-null      float64
 14  unnamed1          0 non-null      float64
dtypes: float64(3), int64(4), object(8)
memory usage: 1.3+ MB
```

- Data statistics

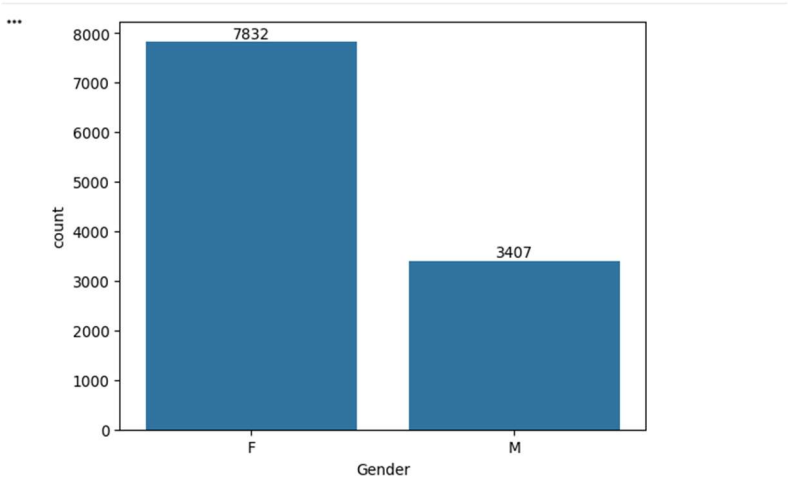
```
df.describe()
```

	User_ID	Age	Marital_Status	Orders	Amount
count	1.123900e+04	11239.000000	11239.000000	11239.000000	11239.000000
mean	1.003004e+06	35.410357	0.420055	2.489634	9453.610858
std	1.716039e+03	12.753866	0.493589	1.114967	5222.355869
min	1.000001e+06	12.000000	0.000000	1.000000	188.000000
25%	1.001492e+06	27.000000	0.000000	2.000000	5443.000000
50%	1.003064e+06	33.000000	0.000000	2.000000	8109.000000
75%	1.004426e+06	43.000000	1.000000	3.000000	12675.000000
max	1.006040e+06	92.000000	1.000000	4.000000	23952.000000

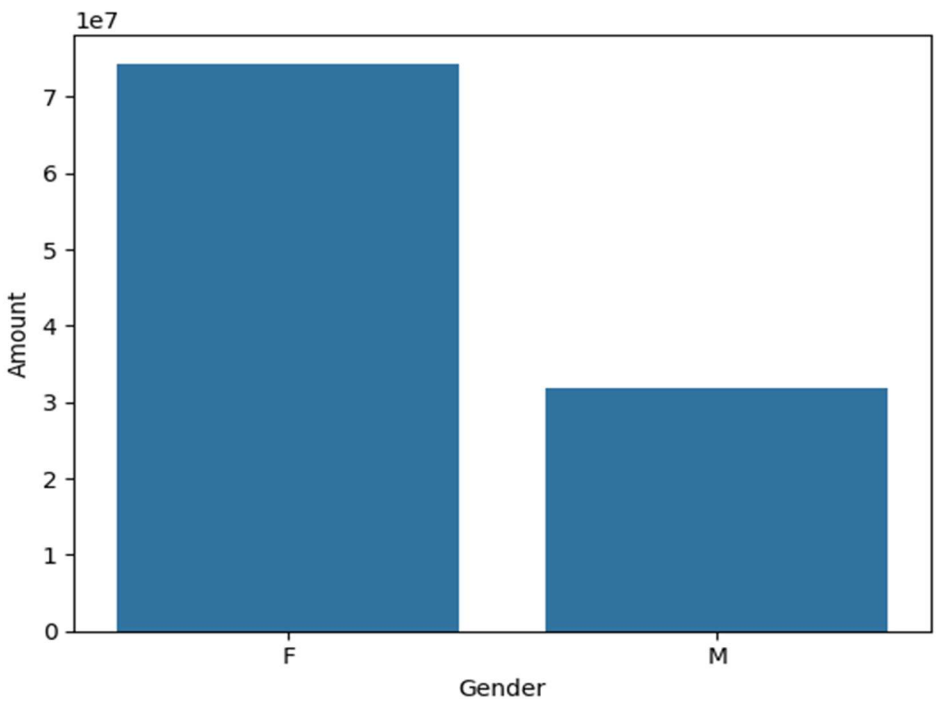
i) I checked the number of Orders that were made as per order category. Order category two had the highest orders followed by one and category three had the least orders.



ii) Number of Purchases per Gender. Female gender had the highest number of purchases compared to men generally. Indicating that the female gender made more orders of products than male. This is a clear indication that the most customers in Diwali sales Store are Female.



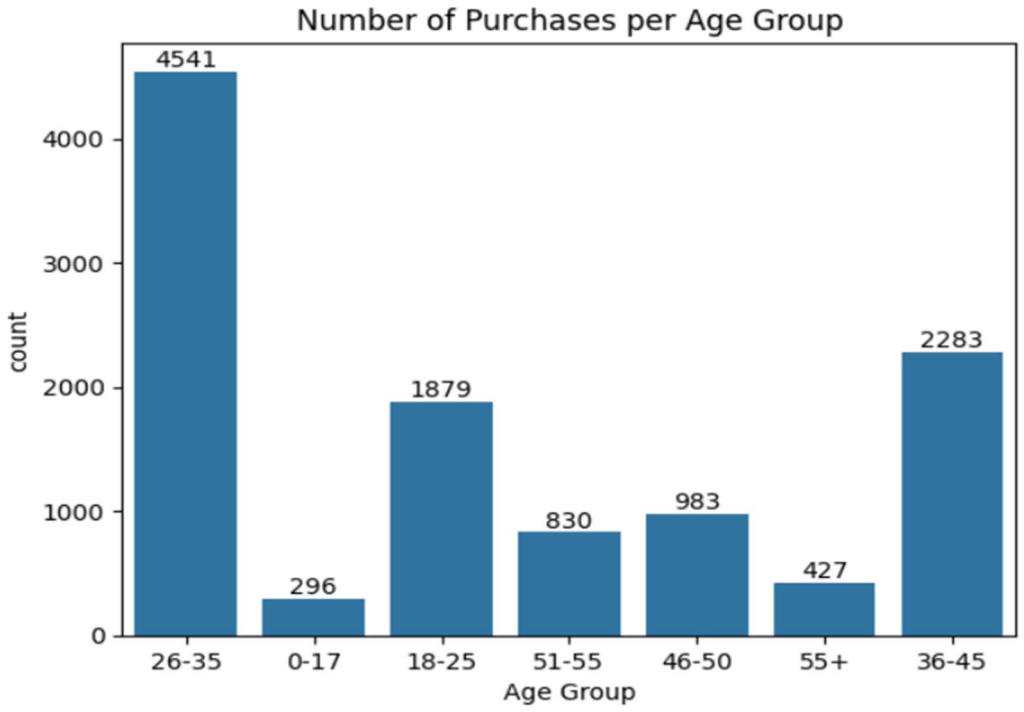
iii) Amount of Revenue generated per Gender; Male and Female. Female generated the highest revenue compared to male. Diwali sales Store need to invest more on products needed by men and improve their mode of service like giving discounts and Bonus to attract more customers.



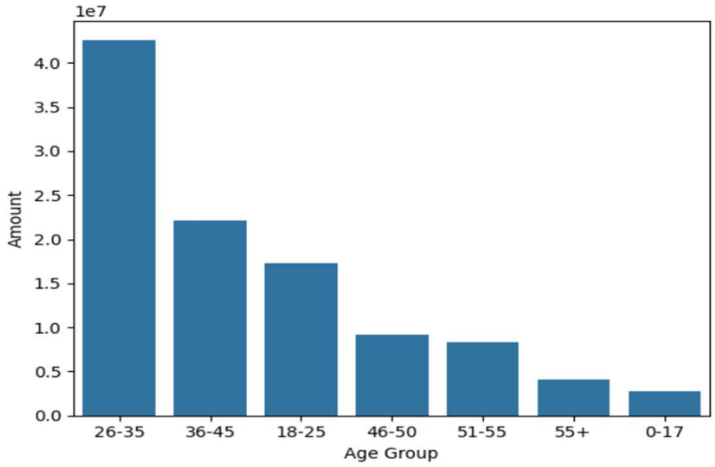
iv)Number of purchases per Age group; The data analyzed shows that people aged between 26-35, (4,541) make more purchases

compared to other Age Groups. This shows that Diwali Store deals on goods that are mostly preferred by youth because age groups, 18-25 and 36-45 are also high, 1879 and 2283 respectively. Diwali sales store should invest more in goods that are more enticing to the youth to increase purchases and earn more revenue.

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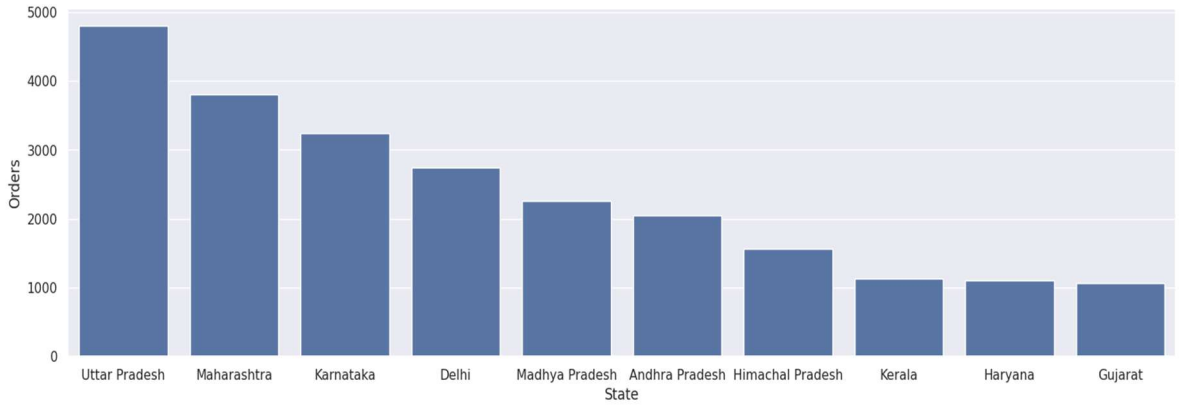
v)Revenue contribution per Age Group:



Interpretation of each age group:

- Age 26–35: This group records the highest amount, approximately 42.5 million. It is the leading age group in the graph.
- Age 36–45: This group has the second-highest amount, approximately 22 million, showing a substantial contribution.
- Age 18–25: This group records approximately 17.3 million, indicating a considerable amount among younger adults.
- Age 46–50: The amount is approximately 9.2 million, considerably lower than that of the younger age groups.
- Age 51–55: This group records approximately 8.3 million, slightly lower than the 46–50 group.
- Age 55+: The amount decreases to approximately 4.1 million.
- Age 0–17: This group records the lowest amount, approximately 2.8 million.

vi) States compared to Orders made:



Top-Performing States:

- **Uttar Pradesh** records the highest order volume with a significant margin, approaching nearly 4,800 orders.
- **Maharashtra** follows in second place with roughly 3,800 orders.
- **Karnataka** and **Delhi** rank third and fourth, generating approximately 3,200 and 2,700 orders respectively.

Mid-Range States:

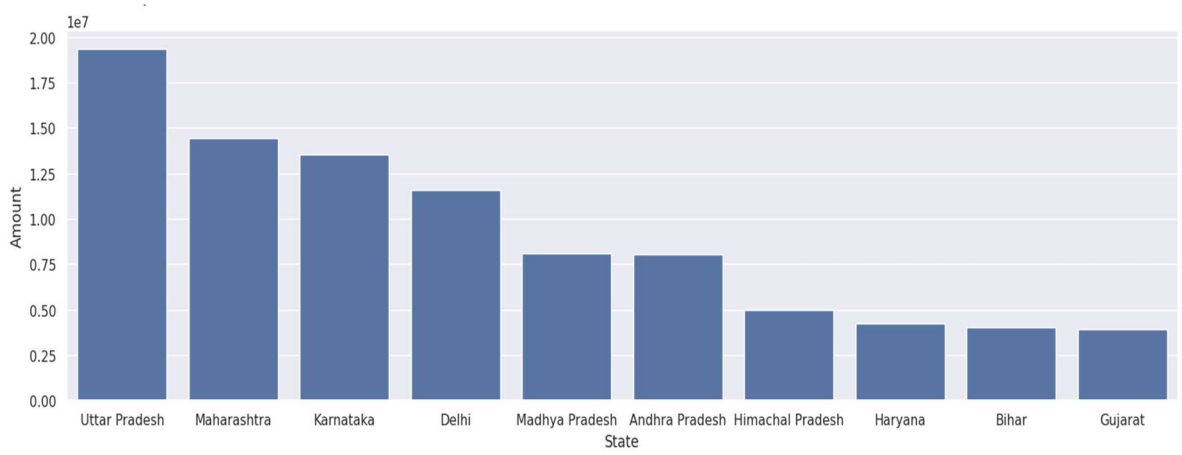
- **Madhya Pradesh** and **Andhra Pradesh** occupy the middle tier, recording about 2,250 and 2,050 orders respectively.

Lower-Tier States:

- **Himachal Pradesh** drops further down to around 1,550 orders.
- **Kerala**, **Haryana**, and **Gujarat** register the lowest order counts on the chart, each clustering closely between 1,000 and 1,100 orders.

It's evident that Uttar Pradesh, Maharashtra and Karnataka offer broad market for Diwali store products contributing to high revenue. Diwali store can expand its business by establishing more branches in in mentioned states above to earn more revenue since those states offer large numbers of customers.

vii) States and Revenue generated (Amount):

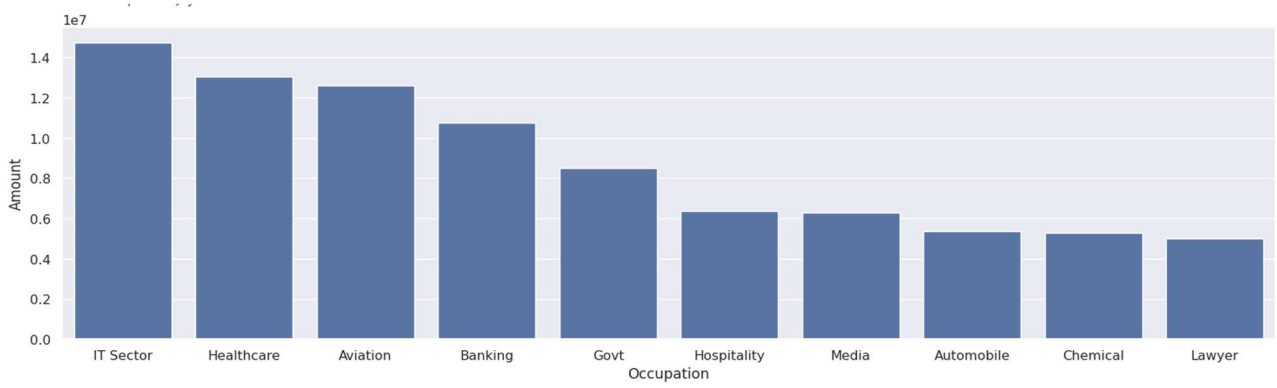


Uttar Pradesh is the dominant market, generating nearly 1.93 million.

Maharashtra follows in second largest market contributing to around 1.44 million.

Karnataka ranks third with around 1.35 million.

viii)Occupation and Amount generated:

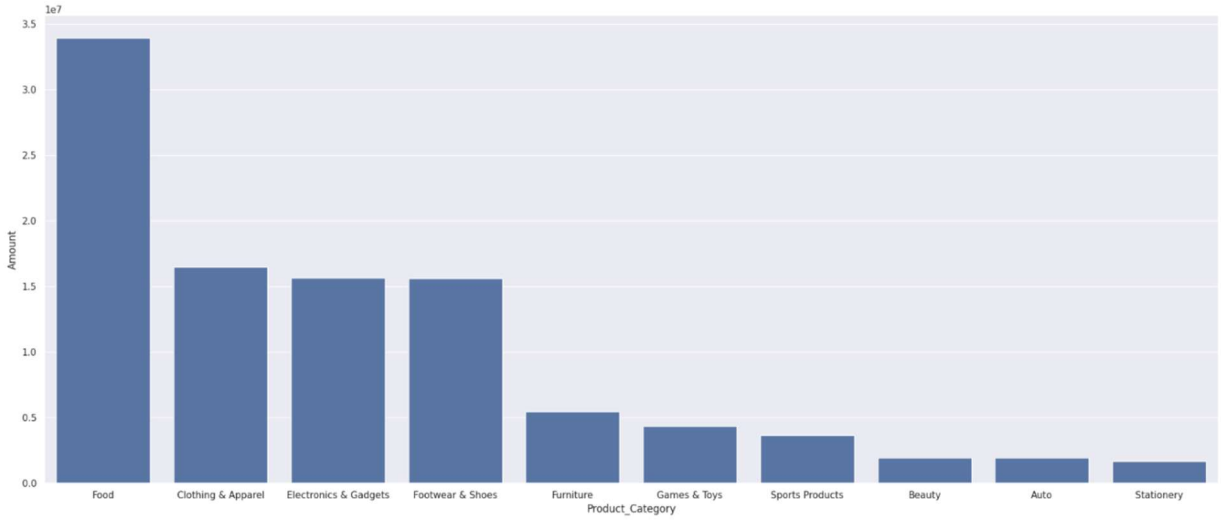


Highest Values:

- **IT Sector:** Leads all industries with the highest total amount, approaching roughly 14.7 million. Diwali sales Store should invest more in IT dominated area since they offer large market.

- **Healthcare:** Ranks second, generating around 13.0 million.
- **Aviation:** Occupies third position contributing about 12.6 million.
- To earn more revenue and expand the business, Diwali Store needs to invest more in IT sector, Healthcare and Aviation industries by establishing or expanding business in these dominated areas in industries.

ix)Product category compared to Amount generated:



Dominant Category:

- **Food:** Dominates in all other categories, reaching nearly 34 million.
- **Clothing & Apparel:** Ranks second at approximately 16.5 million.
- **Electronics & Gadgets:** Generates around 15.6 million.

  Footwear & Shoes: Produces the same Amount with Electronics, around 15.5 million.

Essential consumer goods (Food, Clothing, Electronics, and Footwear) make up much of the total amount, with Food alone generating more than double the amount of any individual secondary category.

Conclusion: Married women, age group 26-35 years from Uttar Pradesh, Maharashtra and Karnataka working in IT, Healthcare and Aviation are more likely to buy products from food, clothing and electronics category. This clearly analyzes and summarizes how Diwali sales Stores should organize its business and make proper strategic decisions by targeting highly profitable areas, dealing in fast moving products and most important consider the states to intensify business operations in.
